

Binary thermodynamic cycles — Solution

X23 (2023) — English translation

A1

0.70 pts

Express the amount of heat Q_- removed from the gas per cycle through T_1 , T_2 , T_4 , R , the amount of gas substance ν and the adiabatic exponent γ .

Solution in TS coordinates.

The figure shows a qualitative diagram of the cycle under consideration in TS coordinates.

Heat is removed from the gas in an isothermal process, so for Q_- we get:

$$Q_- = T_2(S_2 - S_3)$$

where S_2 and S_3 - entropy gasa в точках 2 and 3 соответственно. Since изменение энтропии в адиабатических процессах equals нулю:

$$S_2 - S_3 = S_1 - S_4 = \int_{T_4}^{T_1} \frac{\nu C_p dT}{T} = \frac{\gamma \nu R}{\gamma - 1} \ln \frac{T_1}{T_4}$$

from where we find

Solution in pV or pT coordinates.

The diagram shows a qualitative graph of the process in coordinates pV .

We use the Mendeleev-Clapeyron equation:

$$pV = \nu RT$$

Тепло отводится от gasa в изотермическом процессе, в котором $\delta Q = \delta A$, therefore:

$$\delta Q = p dV = \frac{\nu RT dV}{V} \Rightarrow Q_- = \nu RT_2 \int_{V_3}^{V_2} \frac{dV}{V} = \nu RT_2 \ln \frac{V_2}{V_3}$$

Since процесс 23 изотермический - $V_2/V_3 = p_3/p_2$. Перепишем expression for Q_- :

$$Q_- = \nu RT \ln \frac{p_3}{p_2}$$

For идеального gasa в квазистатическом адиабатическом процессе true equation Пуассона:

$$p \sim T^{\gamma/(\gamma-1)}$$

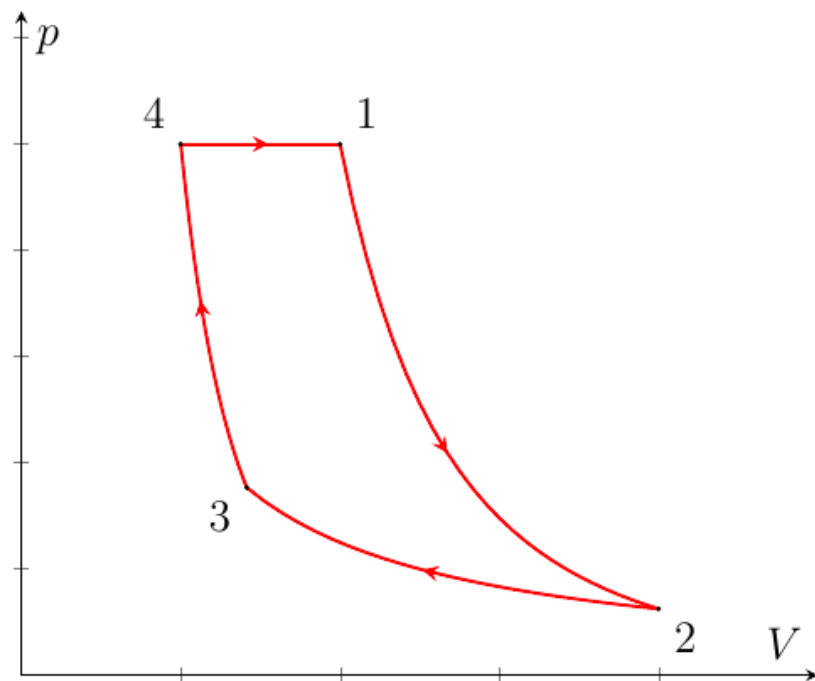
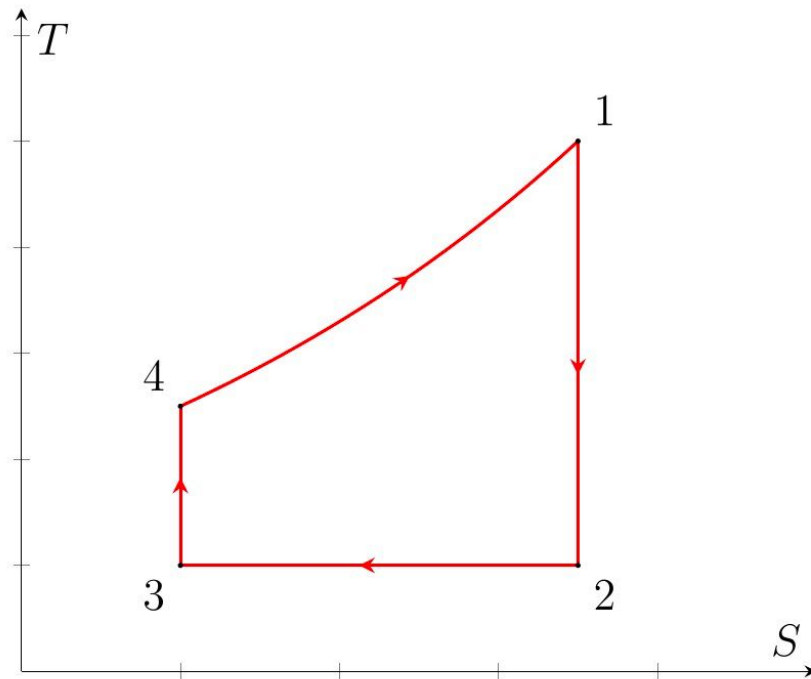
therefore:

$$\begin{cases} \frac{p_1}{p_2} = \left(\frac{T_1}{T_2} \right)^{\gamma/(\gamma-1)} \\ \frac{p_4}{p_3} = \left(\frac{T_4}{T_3} \right)^{\gamma/(\gamma-1)} \end{cases} \Rightarrow \frac{p_3}{p_2} = \left(\frac{T_1}{T_4} \right)^{\gamma/(\gamma-1)}$$

Since $T_3 = T_2$ and $p_4 = p_1$. Finally:

Answer: \textbf{Answer:}

$$Q_- = \frac{\gamma \nu RT_2}{\gamma - 1} \ln \frac{T_1}{T_4}$$



A2

0.80 pts

Express the relation T_1/T_4 in terms of α , β and γ .

Let us write the adiabatic equation for the process 1 – 2 in coordinates pT :

$$p^{1-\gamma}T^\gamma = \text{const} \Rightarrow \frac{T_1}{T_2} = \left(\frac{p_1}{p_2}\right)^{\frac{\gamma-1}{\gamma}}$$

Since $p_1 = p_4$

$$\frac{T_1}{T_2} = \beta^{\frac{\gamma-1}{\gamma}}$$

Also, since $T_2 = T_4/\alpha$:

Answer: \textbf{Answer:}

$$\frac{T_1}{T_4} = \frac{\beta^{\frac{\gamma-1}{\gamma}}}{\alpha}$$

A3

0.50 pts

Obtain an expression for the cycle efficiency η_1 . Express your answer in terms of α , β and γ . Find the numerical value of η_1 for $p_2 = 15 \text{ kPa}$, $p_4 = 60 \text{ kPa}$, $T_2 = 340 \text{ K}$ and $T_4 = 400 \text{ K}$.

Since $Q_+ = C_p(T_1 - T_4)$, we have:

$$\eta_1 = 1 - \frac{C_p T_2 \ln(T_1/T_4)}{C_p(T_1 - T_4)}$$

whence

For α and β we have:

$$\alpha = \frac{20}{17} \quad \beta = 4$$

from where:

Answer: \textbf{Answer:}

$$\eta_1 = 1 - \frac{\ln\left(\beta^{\frac{\gamma-1}{\gamma}}/\alpha\right)}{\beta^{\frac{\gamma-1}{\gamma}} - \alpha}$$

B1

0.50 pts

Find the ratio of the masses of mercury and water $m_{(p)}/m_{(e)}$ participating in the cycle.

Solution in TS coordinates.

The figure shows qualitative diagrams of the considered cycles in TS coordinates.

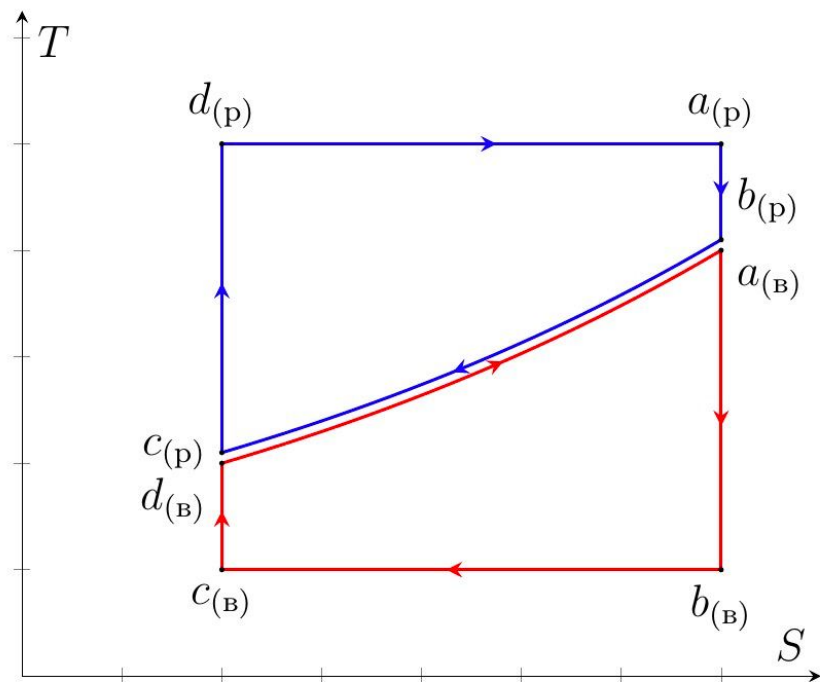
Since the amounts of heat received and given up by water and mercury vapor, respectively, in isobaric polytropic processes are the same at the same temperature changes, the heat capacities in isobaric processes are also the same:

$$C_{p(B)} = \frac{m_{(B)}\gamma_{(B)}R}{\mu_{(B)}(\gamma_{(B)} - 1)} = \frac{m_{(p)}\gamma_{(p)}R}{\mu_{(p)}(\gamma_{(p)} - 1)} = C_{p(p)}$$

from where:

Answer: \textbf{Answer:}

$$\frac{m_{(p)}}{m_{(B)}} = \frac{\mu_{(p)}\gamma_{(p)} - 1}{\mu_{(B)}\gamma_{(p)}} \frac{\gamma_{(B)}}{\gamma_{(B)} - 1} = 17,81$$



B2

0.40 pts

Using graphs of the pressure of saturated vapors of mercury and water versus temperature, estimate the pressure values $p_{c(p)}$ and $p_{c(e)}$.

The required pressures are equal to the pressures of saturated vapors at temperatures t_4 for mercury and t_2 for water. From the graphs we find:

Answer: \textbf{Answer:}

$$p_{c(p)} = 6 \text{ kPa}$$

B3

0.80 pts

Find the mercury and water cycle efficiencies $\eta_{(p)}$ and $\eta_{(e)}$ respectively.

Note: in all paragraphs below it is assumed that $T_i = t_i + T_0$, where $T_0 = 273 \text{ K}$. The change in entropy in isothermal and isobaric processes is the same, therefore:

$$\Delta S = \int_{T_4}^{T_3} \frac{C_p dT}{T} = C_p \ln \frac{T_3}{T_4}$$

Heat, потраченное ртутью в изотермическом процессе, equals:

$$Q_{+(p)} = T_1 \Delta S$$

Heat, отданное водяным варом в изотермическом процессе, equals:

$$Q_{-(B)} = T_2 \Delta S$$

В изобарном процессе водяной varor погашает а ртуть otgives heat Q_p , equalse:

$$Q_p = C_p(T_3 - T_4)$$

therefore answers for $\eta_{(p)}$ and $\eta_{(B)}$ are as follows:

The change in entropy in isothermal and isobaric processes is the same, therefore:

$$\Delta S = \int_{T_4}^{T_3} \frac{C_p dT}{T} = C_p \ln \frac{T_3}{T_4}$$

Heat, absorbed by mercury in an isothermal process, equals:

$$Q_{+(p)} = T_1 \Delta S$$

Heat, released by water vapor in an isothermal process, equals:

$$Q_{-(B)} = T_2 \Delta S$$

In an isobaric process, water vapor releases heat Q_p , equals:

$$Q_p = C_p(T_3 - T_4)$$

therefore answers for $\eta_{(p)}$ and $\eta_{(B)}$ the following:

Answer: \textbf{Answer:}

$$\eta_{(p)} = 1 - \frac{T_3 - T_4}{T_1 \ln(T_3/T_4)} = 0,326$$

B4

0.80 pts

Find the efficiency η_{gn} of the binary gas-steam-water cycle. Compare it with $\eta_{(p)}$ and $\eta_{(B)}$.

Solution in TS coordinates.

Since in an isobaric process gases exchange heat with each other, heat is supplied from the outside to the system participating in the binary cycle only in the mercury isothermal process. Then the efficiency of the binary cycle is the same as that of the Carnot cycle, consisting of two isotherms and two adiabats with temperatures T_1 and T_2 :

$$\eta_{\text{Карно}} = 1 - \frac{T_x}{T_H}$$

Finally:

Answer: \textbf{Answer:}

$$\eta_{gn} = 1 - \frac{T_2}{T_1} = 0,657 > \eta_{(p)}, \eta_{(B)}$$

C1

0.40 pts

Find the temperatures of the system at points 2, 3, 4, 5 and 6.

Since the process 23 is isothermal, and the change in temperature in section 34 can be neglected by condition, the temperatures t_2 , t_3 and t_4 are equal to the temperature of saturated steam at pressure $p_3 = 16$ kPa, i.e.:

The system temperatures at points 5 and 6 correspond to the temperature of saturated steam at pressure $p_4 = 2,8$ MPa, i.e.:

Answer: \textbf{Answer:}

$$t_2 = t_3 = t_4 = t'_1 = 55^\circ\text{C}$$

C2

0.60 pts

Find the amount of heat Q_+ supplied to the system per cycle.

The figure shows a qualitative diagram of the cycle under consideration in TS coordinates.

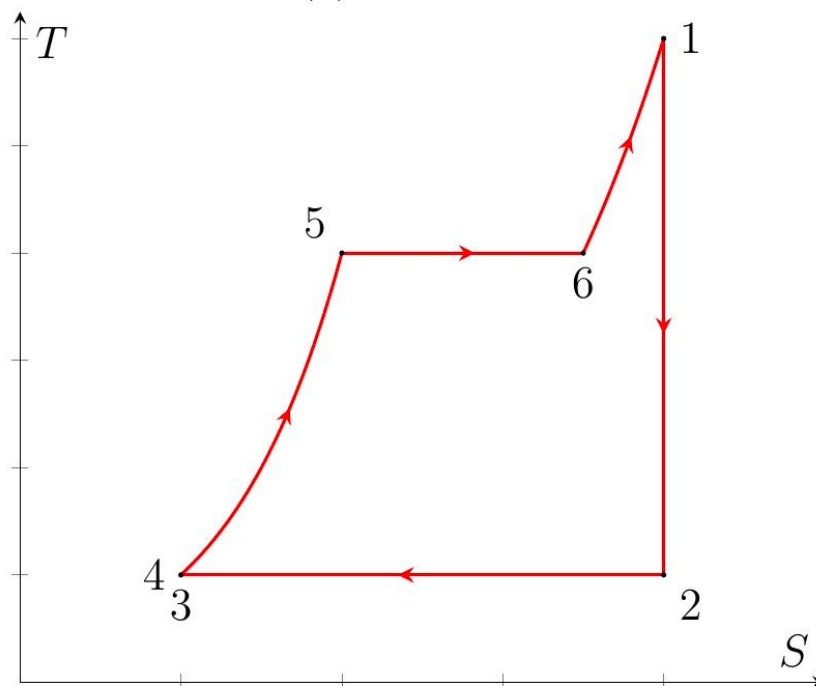
In section 4 – 5 the liquid heats up, in section 5 – 6 it evaporates, and section 6 – 1 corresponds to the overheating of water vapor. Then for Q_+ we have:

$$Q_+ = cm(t_5 - t_4) + L(t_5)m + \frac{\gamma_{(B)}mR(t_1 - t_6)}{\mu_{(B)}(\gamma_{(B)} - 1)}$$

Since $t_6 > 180^\circ\text{C}$ - adiabatic index водяного vapora equals 9/7. Substituting numerical values, we find:

Answer: \textbf{Answer:}

$$Q_+ = 57,35 \text{ MJ}$$



C3

0.80 pts

Find the amount of heat Q_- removed from the system per cycle.

The amount of heat given off per cycle in an isothermal process is equal to:

$$Q_- = T_2(S_2 - S_3)$$

Let us find изменение энтропии другим способом, taking into account that процессы 1 – 2 and 3 – 4 являются адиабатными:

$$S_2 - S_3 = S_5 - S_4 + S_6 - S_5 + S_1 - S_6 = m \left(c \ln \frac{T_5}{T_4} + \frac{L(t_5)}{T_5} + \frac{\gamma_{(B)}R}{\mu_{(B)}(\gamma_{(B)} - 1)} \ln \frac{T_1}{T_6} \right)$$

Thus:

$$Q_- = mT_2 \left(c \ln \frac{T_5}{T_4} + \frac{L(t_5)}{T_5} + \frac{\gamma_{(B)}R}{\mu_{(B)}(\gamma_{(B)} - 1)} \ln \frac{T_1}{T_6} \right)$$

Substituting the numerical values, we find:

Answer: \textbf{Answer:}

$$Q_- = 36,83 \text{ MJ}$$